

available from 12V to 48V with their initial torque varying from 12A to 96A. Typical specifications of various DC pumps and charge controllers available from a vendor are listed in Table II as an example.

Solar panel. The pumps are fitted with motors that receive energy from solar PV arrays. The nominal power of a solar module is expressed in peak



Fig. 2: Drip irrigation (Courtesy: <http://www.urbansustainableliving.com>)

wattage (Wp). The wattage of solar panels depends on the requirement and the motors used. In India, solar PV arrays having capacity in the range of 200W to 5kWp are recommended by the government. You may refer to information on solar PV arrays and motor pump sets for solar PV water pumping systems (2013-14) given by the Ministry of New and Renewable Energy, under Jawaharlal Nehru National Solar Mission, from the link: http://mnre.gov.in/file-manager/UserFiles/technical-specification_spwps_2013_14.pdf

Many vendors offer installation services at a cost. But if you are planning to install it yourself, you can refer many live video clips available on the Internet. One such link is available at <http://www.solarpumps.com/>

Types of motor pumps

Solar pumps fall into two major categories, surface and submersible, depending on their construction and applications.

Surface pump. Surface pump is usually installed above the water level which requires a suction lift. It is used to get discharge from an open well, pond or river where high flow and small head (height to which the water needs to be lifted to) are required. There are many different types of pumps available, and DC surface pumps with total lift up to 21m (70 feet) are easily available in India.

Submersible

pump. This pump is basically a hermetically sealed motor. The whole assembly of submersible pump is submerged in the water to be pumped up. It is generally used for pumping water from small-bore wells, and is designed to fit inside the drilled bore. The well may be anywhere from a few metres to 550 metres deep. But the deeper the well, the more powerful and expensive would be the system. The 0.25HP to 3HP pumps are easily available in the market. DC submersible pumps with total lift up to 92m are quite common in India.

Submersible pumps have the ability to lift water to a greater height. So these pumps are also used for water transfer applications from a canal to storage tanks or from an underground storage tank to an overhead tank.

Pump to buy

Before buying a pump you should know the depth at which water is available, daily water requirement and type of irrigation system to be installed. You may also need to know the water flow rate and the total head of the pump. You should buy a pump having the best efficiency. The best combination of head and flow rate at the least energy consumption gives the best pump efficiency.

The performance graph of 1.5HP motor pump available from a vendor is shown in Fig. 3 as an example. To interpret the graph and the table, let us say you have an input power of 400 watts from the PV array and you want to lift the water to a height of 40 metres at a flow rate of 12 litres per minute. As per Fig. 3, you would need a 1.5HP pump-set with pressure of 56.88psi or 392 kilopascals (392kPa).

Next step is to decide whether you should go for AC pump or DC pump. Though a DC pump can be run directly from a solar PV array, it has very limited range as compared to AC pump. The DC pump costs more than AC pump with the same category and the same head. But if a DC pump is available in your local market that meets your requirement it can be worth buy-

TABLE I
Typical AC Pump and Controller Specifications

AC pump HP	Volts	Controller	Solar panel (watt-peak)	Battery capacity
1HP	60	4kVA	1500Wp	100Ah - 5 Nos.
1.5HP	72	5kVA	1800Wp	100Ah - 6 Nos.
2HP	96	6kVA	2400Wp	100Ah - 8 Nos.
2.5HP	96	6kVA	2400Wp	120Ah - 8 Nos.

Courtesy: www.solarenergyproducts.co.in

TABLE II
DC Pumps with Charge Controllers

DC pump	0.125HP	0.25HP	0.5HP	1HP	2HP
Volts	24V	24V	24V	48V	48V
No-Load Current	2A	4A	8A	8A	16A
Full-Load Current	4A	8A	15A	16A	32A
Initial Torque	12A	24A	45A	48A	96A
Suction	1.83m (6 feet)	7.62m (25 feet)	16.46m (54 feet)	76.2m (250 feet)	152.4m (500 feet)
Delivery or Flow Rate (litres per hour or minute)	500-lph	8-lpm	24-lpm	35-lpm	70-lpm
Panel Capacity	200Wp	500Wp	800Wp	1500Wp	3000Wp
Battery Capacity	50Ah - 2 Nos.	150Ah - 2 Nos.	200Ah - 2 Nos.	200Ah - 4 Nos.	200Ah - 8 Nos.
Charge Controller Rating	24V, 40A	24V, 40A	24V, 40A	48V, 40A	48V, 40A - 2 Nos.